

University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering

SPCE 5065

HOMEWORK #5

DUE: Friday 31 July 2026

Instructions:

- Include your references at the end of your homework in APA format.
<https://www.aiaa.org/publications/journals/reference-style-and-format>
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions

1. For **each** the current events presentations this week

- a) Summarize the presentation
- b) Describe something you learned from it
- c) Write one question you have left about the presentation

2. Kinetic energies of space objects can be very large. **Clearly state your assumptions for this problem.**

- a. Plot the kinetic energy (in Joules) versus diameter of a particle on a log-log scale. Assume the particle is at the same altitude as the ISS and that its density is 1g/cm^3 .
- b. How does this compare to the Kinetic energy of a small (5 kg), medium (50 kg) and large satellite (100 kg) in the same orbit?
- c. What type of satellites are most susceptible to damage due to impacts with man-made space objects?

3. A 1.0-g piece of space debris strikes a spacecraft in low Earth orbit. Its kinetic energy is said to be equivalent to the gravitational potential energy lost by a 2.0-kg bowling ball dropped from a height of 100 m near Earth's surface

- a. Calculate the impact speed of the debris. Clearly identify all assumptions.
- b. Micrometeoroids can encounter spacecraft at much greater relative speeds. Calculate the kinetic energy of the same 1.0-g particle if its impact speed were 20 km/s. Express the result in joules and as an equivalent height from which the 2.0-kg bowling ball would need to fall.
- c. How does this compare to the kinetic energy of the space debris?

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d. Which poses the greater threat? Be sure to consider the likelihood of impact (although you don't need to quantify it).

4. Research an orbital debris modelling program, either mentioned in class or find another from a reliable source. Describe who publishes and maintains it, its key features, and how it works. Summarize its predicted effects of space debris on the ISS.

5. Compare the likelihood of satellites in typical orbits of being impacted by micrometeoroids. For the purposes of this exercise, assume the altitude of the atmosphere is 100 km.

a. Plot the flux density for micrometeoroids with masses ranging from 10^{-5} g to 10 g for the ISS, GPS, and a GEO satellite on the same plot.

b. How much average time (in years) will there be between events (collisions) with probabilities greater than 0.01% for each orbit for micrometeoroids of 0.1 cm, 1 cm, and 10 cm? Assume the satellite has an area of ten m².

c. What can you conclude about the likelihood of damage to satellites with an expected mission life of 10 years?

6. Find a mission or incident, other than the Chinese Fengyun-1C ASAT, which was a violation of one of the UN guidelines published in 2010. What action could have been taken? What action was taken?

7. Describe three of the technology solutions to help reduce the harmful effects of space debris. Explain how it might be implemented and provide one example.

GRADUATE ONLY:

8. Compare the US Orbital Debris Mitigation Standard Practices policy with another country's policy.

a. Summarize the key provisions of each.

b. What are the key similarities and differences?

c. Research how/if they are enforced.

9. A spacecraft with a Whipple shield is required to survive the impact of a meteoroid with a density of 1.6 g/cm³, diameter of 1 cm, and velocity of 80 km/s. The shield and spacecraft are made of aluminum 7075 T6 with a yield stress of 65 ksi. Determine and plot the shield and wall thickness as a function of offset distance of 1 to 30 cm. Clearly state your assumptions.